Joshua L. Flynn

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Massachusetts Institute of Technology
Department of Mathematics
Cambridge, MA

EMPLOYMENT

- **NSF Postdoctoral Fellow**, Massachusetts Institute of Technology (2024–present)
- **CRM-ISM Postdoctoral Fellow**, McGill University (2022–2024)

MATHEMATICAL RESEARCH INTERESTS

Geometric analysis and non-Euclidean geometric measure theory, with particular interest in sharp functional and geometric inequalities in non-Euclidean settings.

AI-ASSISTED MATHEMATICAL RESEARCH

- Develop and employ multi-agent AI systems for mathematical theory development.
- Develop and design serialization of mathematical works into structured data sets amenable to various data-analysis techniques.
- Apply these systems to problems in geometric analysis and non-Euclidean geometric measure theory.

EDUCATION

University of Connecticut

- Ph.D. in Mathematics (2022)
- Dissertation: *Sharp Geometric Inequalities on Noncompact Rank One Symmetric Spaces and Their Boundaries*
- Advisor: Guozhen Lu

University of Washington

- B.Sc. in Mathematics (2015)

Tacoma Community College

- Attended (2009–2012)

PAPERS

1. Flynn, J. L., & Tejeda, I. *Parabolic Varifolds v0.1*. **Preprint** (2026).
2. Flynn, J., Lu, G., & Yang, Q. *Conformally covariant boundary operators and sharp higher order CR Sobolev trace inequalities on the Siegel domain and complex ball*. **Journal für die reine und angewandte Mathematik** (2025).
3. Flynn, J., Lam, N., & Lu, G. *L^p-Hardy identities and inequalities with respect to the distance and mean distance to the boundary*. **Calculus of Variations and Partial Differential Equations** (2025).
4. Flynn, J., & Vétois, J. *Liouville-type result for the CR Yamabe equation in the Heisenberg group*. **Annali della Scuola Normale Superiore di Pisa — Classe di Scienze** (2024).
5. Cazacu, C., Flynn, J., Lam, N., & Lu, G. *Caffarelli-Kohn-Nirenberg identities, inequalities and their stabilities*. **Journal de Mathématiques Pures et Appliquées** (2024).
6. Flynn, J., Lu, G., & Yang, Q. *Sharp Hardy-Sobolev-Maz'ya, Adams and Hardy-Adams inequalities on quaternionic hyperbolic spaces and on the Cayley hyperbolic plane*. **Revista Matemática Iberoamericana** (2024).
7. Flynn, J., Lu, G., & Yang, Q. *Conformally covariant boundary operators and sharp higher order Sobolev trace inequalities on Poincaré-Einstein manifolds*. **Preprint** (2023).
8. Do, A. X., Flynn, J., Lam, N., & Lu, G. *L^p-Caffarelli-Kohn-Nirenberg inequalities and their stabilities*. **Preprint** (2023).
9. Flynn, J., & Reznikov, J. *General Conformally Induced Mean Curvature Flow*. **Preprint** (2023).
10. Cazacu, C., Flynn, J., & Lam, N. *Caffarelli-Kohn-Nirenberg inequalities for curl-free vector fields and second order derivatives*. **Calculus of Variations and Partial Differential Equations** (2023).
11. Flynn, J., Lam, N., Lu, G., & Mazumdar, S. *Hardy's identities and inequalities on Cartan-Hadamard manifolds*. **The Journal of Geometric Analysis** (2023).
12. Flynn, J., Lam, N., & Lu, G. *Hardy-Poincaré-Sobolev type inequalities on hyperbolic spaces and related Riemannian manifolds*. **Journal of Functional Analysis** (2022).
13. Cazacu, C., Flynn, J., & Lam, N. *Sharp second order uncertainty principles*. **Journal of Functional Analysis** (2022).
14. Burchard, A., Flynn, J., Lu, G., & Shaw, M.-C. *Extendability and the $\bar{\partial}$ operator on the Hartogs triangle*. **Mathematische Zeitschrift** (2022).
15. Cazacu, C., Flynn, J., & Lam, N. *Short proofs of refined sharp Caffarelli-Kohn-Nirenberg inequalities*. **Journal of Differential Equations** (2021).
16. Flynn, J., Lam, N., & Lu, G. *Sharp Hardy identities and inequalities on Carnot groups*. **Advanced Nonlinear Studies** (2021).

17. Flynn, J. *Sharp L^2-Caffarelli–Kohn–Nirenberg inequalities for Grushin vector fields*. **Nonlinear Analysis** (2020).
18. Flynn, J. *Sharp Caffarelli–Kohn–Nirenberg-type inequalities on Carnot groups*. **Advanced Nonlinear Studies** (2020).

RESEARCH SUPERVISION

- **MIT UROP Supervisor** (2026–present)
  - Supervising an undergraduate research project on cellular automata, fractal geometry, and related sparse-type operators.
- **McGill/MIT Undergraduate Research Project Co-Supervisor** (Summer 2025)
- **McGill Department Undergraduate Research Award Co-Supervisor** (Summer 2024)
- **McGill Undergraduate Harmonic Analysis Reading Group Supervisor** (Summer 2024)
- **Directed Reading Program Volunteer**, University of Connecticut (2021)
  - Mentored an undergraduate mathematics student on the finite element method and MATLAB.

TEACHING EXPERIENCE

- **Instructor**, 18.100A Real Analysis, MIT (Fall 2026)
- **Teaching Assistant**, 18.02 Multivariable Calculus, MIT (Spring 2026)
- **Teaching Assistant**, 18.03 Differential Equations, MIT (Fall 2025)
- **Instructor**, ECSE 507 Optimization and Optimal Control, McGill (Winter 2024)
- **Instructor**, ECSE 500 Mathematical Foundations of Systems, McGill (Fall 2023)
- **Teaching Assistant**, Calculus I (Math 1131Q), UConn (Spring 2022)
- **Instructor**, Technical Writing in Mathematics (Math 2705W), UConn (Fall 2021)
- **Teaching Assistant**, Elementary Differential Equations (Math 2410Q), UConn (Spring 2021)
- **Teaching Assistant**, Elementary Differential Equations (Math 2410Q), UConn (Fall 2020)
- **Instructor**, Elementary Differential Equations (Math 2410Q), UConn (Spring 2020)
- **Instructor**, Calculus for Business and Economics (Math 1071Q), UConn (Fall 2019)
- **Instructor**, Elementary Differential Equations (Math 2410Q), UConn (Spring 2019)
- **Teaching Assistant**, Calculus I (Math 1131Q), UConn (Fall 2018)
- **Instructor**, Elementary Differential Equations (Math 2410Q), UConn (Spring 2018)
- **Teaching Assistant**, Calculus I (Math 1131Q), UConn (Fall 2017)
- **Instructor**, Calculus for Business and Economics (Math 1071Q), UConn (Spring 2017)
- **Instructor**, Calculus for Business and Economics (Math 1071Q), UConn (Fall 2016)
- **Instructor**, Calculus for Business and Economics (Math 1071Q), UConn (Spring 2016)
- **Instructor**, Problem Solving (Math 1020Q), UConn (Fall 2015)

PROFESSIONAL DEVELOPMENT

- **MIT Grant Writing Training Certificate** (expected 2027)
- **MIT Research Mentoring Certificate** (completed June 2026)
- **MIT Kaufman Teaching Certificate Program** (completed May 2026)

FELLOWSHIPS AND AWARDS

- **NSF Mathematical Sciences Postdoctoral Research Fellowship** (2024–present)
- **CRM-ISM Postdoctoral Fellowship** (2022–2024)
- **Michael Neumann Dissertation Award** (2022)
- **Conference Participation Award** (2021)
- **Summer Doctoral Dissertation Fellowship** (2021)
- **Predoctoral Fellowship** (2020)
- **Provost Excellence in Teaching** (2019)

TALKS

Seminars and Colloquia

- GAP Seminar, Pennsylvania State University, State College, PA (November 2026; scheduled).
- *Anisotropic Varifolds*. Colloquium, University of New Mexico, Albuquerque, NM (January 2026).
- *Anisotropic Varifolds*. Geometric Analysis Seminar, McGill University, Montréal, QC (November 2025).
- *A Theory of Anisotropic Varifolds*. Geometry & Topology Seminar, Caltech, Pasadena, CA (June 2025).
- *Conformal Boundary Operators, Extension Theorems and Trace Inequalities*. Geometric Analysis Seminar, McGill University, Montréal, QC (March 2024).
- *Geometric Flows, Applications and the Isoperimetric Problem*. Colloquium, Oregon State University, Corvallis, OR (March 2024).
- *The Isoperimetric Problem and Related Mean Curvature Type Flows*. PDE, Complex Analysis and Differential Geometry Seminar, University of Notre Dame, South Bend, IN (October 2023).
- *Sharp Hardy-Sobolev-Maz'ya Inequalities for Noncompact Rank One Symmetric Spaces*. Geometric Analysis Seminar, McGill University, Montréal, QC (November 2022).
- *Sharp Uncertainty Principles for Physical Vector Fields and Second-Order Derivatives*. Geometric and Functional Inequalities and Applications, international virtual seminar (October 2022).
- *Helgason-Fourier Analysis on the Hyperbolic Spaces*. Séminaire Analyse Harmonique, Centre de Recherches Mathématiques, Montréal, QC (October 2022).
- *Some Sharp Uncertainty Principles and Related Geometric Inequalities*. Ghent Methusalem Junior Seminar, virtual (May 2022).
- *Hardy-Sobolev-Maz'ya Inequalities on Siegel Domains*. Mathematics Seminar, Bridgewater State University, virtual (March 2022).
- *Sharp Caffarelli-Kohn-Nirenberg Inequalities for Grushin Vector Fields and Iwasawa Groups*. Geometric and Functional Inequalities and Applications, international virtual seminar (November 2020).

Conferences

- *Higher-Order Trace Inequalities on Poincaré-Einstein Manifolds*. AMS Special Session on Geometric and Functional Inequalities: Sharp Constants, Extremals, Stability and Their Applications, Boston, MA (March 2026).
- *Liouville-Type Result for the CR Yamabe Equation in the Heisenberg Group*. Workshop in PDE and Applied Math in the Northeast Region, Storrs, CT (March 2026).
- *Higher-Order Boundary Operators and Trace Inequalities on the Siegel Domain and Complex Ball*. CMS Special Session on Harmonic Analysis & PDE, Toronto, ON (December 2025).
- *Mean Curvature Type Flows and the Isoperimetric Problem*. AMS Special Session on Sharp Geometric Inequalities and Their Stability, PDEs and Applications, Hartford, CT (April 2025).
- *Conformal Boundary Operators, Extension Theorems and Trace Inequalities on Poincaré-Einstein Manifolds*. Analysis of Geometric Singularities, Centre de Recherches Mathématiques, Montréal, QC (May 2024).
- *Liouville-Type Result for the CR Yamabe Equation in the Heisenberg Group*. CMS Special Session on Harmonic Analysis & PDE, Montréal, QC (December 2023).
- *The Isoperimetric Problem and Related Mean Curvature Type Flows*. CMS Special Session on Geometric Functional Analysis: Analytic, Discrete, and Probabilistic Aspects, Montréal, QC (December 2023).
- *General Conformally Induced Mean Curvature Flow*. AMS Southeastern Sectional Meeting Special Session on Topics in Harmonic Analysis and Partial Differential Equations, University of South Alabama, Mobile, AL (October 2023).
- *Hardy Inequalities and Mean Convex Domains*. CMS Special Session on Interplay Between Analysis and Convexity, University of Ottawa, Ottawa, ON (June 2023).
- *The Caffarelli-Kohn-Nirenberg Identities, Inequalities and Their Stabilities*. AMS Spring Central Sectional Meeting Special Session on Geometric and Analytic Methods in PDE, University of Cincinnati, Cincinnati, OH (April 2023).
- *Sharp Hardy Integral Identities and Inequalities on Domains*. AMS Special Session on Advances and Applications in Integral and Differential Equations, Joint Mathematics Meeting, Boston, MA (January 2023).
- *Sharp Second-Order Uncertainty Principles and Related Inequalities for Vector Fields*. AMS Special Session on Advances in Partial Differential Equations, Numerical Analysis, and Their Applications, Joint Mathematics Meeting, Boston, MA (January 2023).
- *Helgason-Fourier Analysis and Sharp Geometric Inequalities on the Rank One Symmetric Spaces*. CMS Special Session on Approximation Theory, Function Spaces and Harmonic Analysis, University of Toronto, Toronto, ON (December 2022).
- *Hardy-Sobolev-Maz'ya and Adams Inequalities for the Quaternionic Hyperbolic Spaces*. AMS Spring Western Sectional Meeting Special Session on Geometric and Functional Inequalities and Applications to PDEs, virtual (May 2022).
- *Sharp Hardy-Sobolev-Maz'ya, Adams and Hardy-Adams Inequalities on Quaternionic Hyperbolic Spaces and the Cayley Hyperbolic Plane*. Joint Mathematics Meeting, virtual (April 2022).
- *Sharp Hardy-Sobolev-Maz'ya, Adams and Hardy-Adams Inequalities on Quaternionic Hyperbolic Spaces and the Cayley Hyperbolic Plane*. 17th Prairie Analysis Seminar, Kansas State University, virtual (November 2021).
- *Sharp Caffarelli-Kohn-Nirenberg Inequalities for the Grushin Operator and Iwasawa Groups*. Annual New York State Regional Graduate Mathematics Conference, Syracuse University, virtual (April 2021).
- *Hardy Identities on Domains*. AMS Spring Eastern Sectional Meeting Special Session on Metric Techniques in Analysis, virtual (March 2021).

- *Sharp Caffarelli-Kohn-Nirenberg Inequalities for Grushin Vector Fields and Iwasawa Groups*. Tenth Ohio River Analysis Meeting, University of Kentucky, virtual (March 2021).
- *Hardy-Sobolev-Maz'ya and Adams Inequalities on Quaternionic Hyperbolic Spaces*. AMS Spring Eastern Sectional Meeting Special Session on Geometric and Functional Inequalities and Nonlinear Partial Differential Equations, virtual (March 2021).
- *Algebraic Aspects of Dunkl Theory*. Binghamton University Graduate Conference in Algebra and Topology, virtual (November 2020).
- *Sharp L^2-Caffarelli-Kohn-Nirenberg-Type Inequalities for the Grushin Operator and Iwasawa Groups*. AMS Fall Central Sectional Meeting Special Session on Geometric Inequalities and Nonlinear Partial Differential Equations, virtual (September 2020).
- *Sharp Caffarelli-Kohn-Nirenberg Type Inequalities on Iwasawa Groups*. Joint Mathematics Meeting, Denver, CO (January 2020).
- *Some Geometric Inequalities by Algebraic Trickery*. Mathematics Continued Conference, University of Connecticut, Storrs, CT (November 2019).

CONFERENCE AND SEMINAR ORGANIZATION

- CMS Special Session on *Nonlinear and Geometric PDEs*, Montréal, QC, co-organizer (Dec 11–14, 2026).
- AMS Special Session on *Topics in Geometric Analysis and PDEs*, Boston, MA, co-organizer (Mar 28, 2026).
- AMS Special Session on *Research in Analysis and PDEs by Early Career Mathematicians*, virtual, co-organizer (Oct 25–26, 2025).
- AMS Special Session on *Sharp Geometric Inequalities and their Stability, PDEs and Applications*, Hartford, CT, co-organizer (Apr 5–6, 2025).
- CMS Special Session on *Geometric Analysis & PDE*, Vancouver, BC, co-organizer (Nov 29–Dec 2, 2024).
- *Postdocs at CRM Seminar*, CRM/UdeM, co-organizer (2023–2024).
- *Geometric Analysis Seminar*, McGill University, co-organizer (2022–2024).
- JMM Special Session on *Advances in Partial Differential Equations, Numerical Analysis, and their Applications*, Boston, MA, co-organizer (Jan 4–7, 2023).
- AMS Special Session on *Geometric and Functional Inequalities and Applications to PDEs*, Spring Western Sectional Meeting, co-organizer (May 14–15, 2022).
- AMS Special Session on *Geometric and Functional Inequalities and Nonlinear Partial Differential Equations*, Spring Eastern Sectional Meeting, co-organizer (Mar 20–21, 2021).
- AMS Special Session on *Geometric Inequalities and Nonlinear Partial Differential Equations*, Fall Central Sectional Meeting, co-organizer (Sep 12–13, 2020).
- *Geometric and Functional Inequalities and Applications*, international virtual seminar, co-organizer (2020–present). https://researchseminars.org/seminar/GeomInequAndPDEs

SERVICE AND MENTORING

- **Association for Women in Mathematics Mentor Network** (2025–present)
- **MIT School of Science Postdoc Representative** (2025–present)
- **CRM Equity-Diversity-Inclusion Committee Member** (2022–2024)
- **TA Mentor Network Volunteer**, University of Connecticut (2017–2022)
  - Mentored incoming graduate students in the mathematics Ph.D. program at UConn.
- **Preliminary Exam Review Instructor**, University of Connecticut (Summer 2020)
  - Held weekly review sessions to help graduate students prepare for the analysis preliminary exam.

REFEREE SERVICE

Journal of Differential Geometry, *Mathematische Annalen*, *Transactions of the American Mathematical Society*, *The Journal of Geometric Analysis*, *Advances in Calculus of Variations*, *Nonlinear Analysis*, *Forum Mathematicum*, *Potential Analysis*, *Advances in Nonlinear Analysis*, *Acta Mathematica Sinica*, *Communications on Pure and Applied Analysis*, *Archiv der Mathematik*, *Rendiconti del Seminario Matematico della Università di Padova*, and *Acta Universitatis Sapientiae, Mathematica*.

TECHNICAL SKILLS

Programming & Computing	Python, Java, MATLAB, Git, Linux
Cloud Computing	AWS
Agentic Mathematics Systems	Danus, Rethlas
AI Development Tools	Codex, Claude Code
Formalization	Lean, Archon (Lean formalizer)